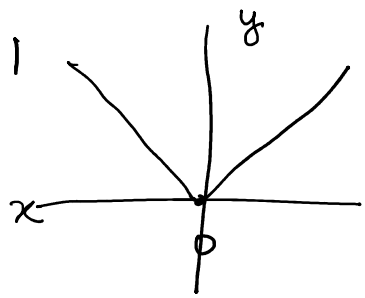
LECTURE 11

03 October 2023 19:27

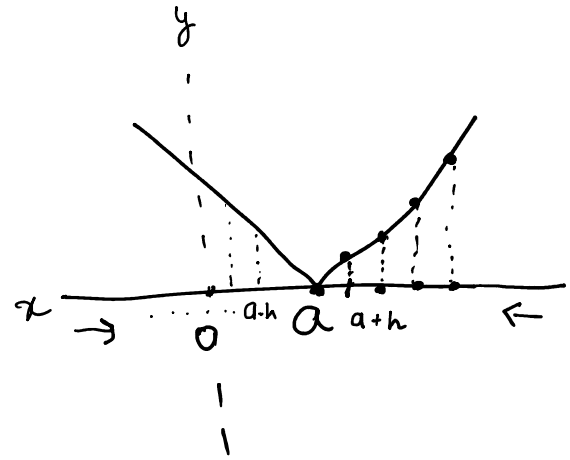
$f : A \to B$

$A \downarrow$ domain, $B \downarrow$ co-domain

$f(x) = |x|$



$f(x) = |x - a|$



$RHL = f(0+0) = \lim_{h \to 0} f(0+h)$

$= \lim_{h \to 0} |h|$

$= \lim_{h \to 0} h = 0$

$RHL = f(a+0) = \lim_{h \to 0} f(a+h) = l$

$LHL = f(a-0) = \lim_{h \to 0} f(a-h) = l$

$LHL = f(0-0)$

$= \lim_{h \to 0} f(0-h)$

$= \lim_{h \to 0} h = 0$

$RHL = LHL = 0$

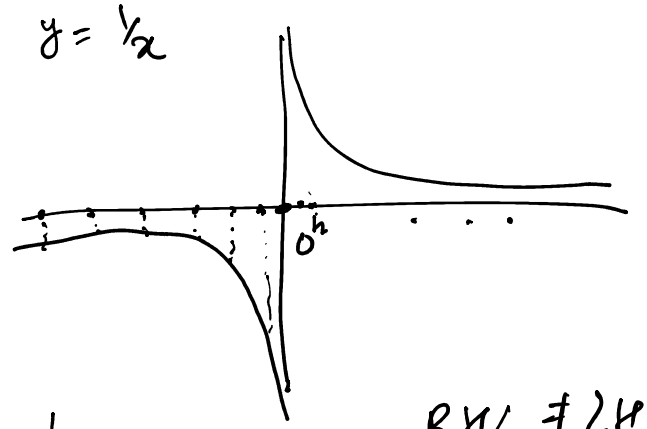
$RHL = LHL = l$

$\lim_{x \to a} f(x) = l$

$\boxed{\lim_{x \to 0} |x| = 0}$

$f(x) = 1/x$

$y = 1/x$



$RHL = f(0+0) = \lim_{h \to 0} f(0+h)$

$= \lim_{h \to 0} \frac{1}{h} = \infty$

$LHL = f(0-0) = \lim_{h \to 0} f(0-h) = \lim_{h \to 0} \frac{-1}{h} = -\infty$

$RHL \neq LHL$

limit doesn't exist

$LHL = f(0-0) = \lim_{h\to 0} f(0-h) = \lim_{h\to 0} -\frac{1}{h} = -\infty$ RHL ≠ LHL limit doesn't exist

$f(x) = x \sin\frac{1}{x}$ $\lim_{x\to 0} f(x) = ??$ 0

$RHL = f(0+0) = \lim_{h\to 0} f(0+h) = \lim_{h\to 0} (0+h) \sin\frac{1}{0+h}$

$= \lim_{h\to 0} h \sin\frac{1}{h} = 0$

$LHL = f(0-0) = \lim_{h\to 0} f(0-h) = \lim_{h\to 0} (0-h) \sin\frac{1}{0-h}$

$= \lim_{h\to 0} +h \sin\frac{1}{h} = 0$ $RHL = LHL = 0$

$f(x) = \begin{cases} 3x-2 & x<1 \\ 4x^2-3x & x\geq 1 \end{cases}$ $\lim_{x\to 1} f(x) = ??$ 1

$RHL = f(1+0) = \lim_{h\to 0} f(1+h) = \lim_{h\to 0} 4(1+h)^2 - 3(1+h)$

$= \lim_{h\to 0} 4(1+0)^2 - 3(1+0) = 4-3 = 1$

$LHL = f(1-0) = \lim_{h\to 0} f(1-h) = \lim_{h\to 0} 3(1-h)-2 = 3(1-0)-2$
$= 1$

$RHL = LHL = 1$

$f(x) = \sin\frac{1}{x}$ $\lim f(x) = ??$ doesn't exist.

$f(x) = \sin \frac{1}{x-1}$ $\qquad \lim_{x \to 1} f(x) = ??$ doesn't exist.

$RHL = f(1+0) = \lim_{h \to 0} f(1+h) = \lim_{h \to 0} \sin \frac{1}{1+h-1} = \lim_{h \to 0} \sin \frac{1}{h}$

$= \text{osci. } -1 \text{ \& } 1$

$LHL = f(1-0) = \lim_{h \to 0} f(1-h) = \lim_{h \to 0} \sin \frac{1}{1-h-1} = \lim_{h \to 0} \sin \frac{1}{-h}$

$= - \lim_{h \to 0} \sin \frac{1}{h} \Rightarrow \text{osci. } -1 \text{ \& } 1$

$RHL \neq LHL \Rightarrow$ Both don't exist.

$f(x) = [x]$ $\qquad \lim_{x \to 1} [x] = ??$ does not exist.

$RHL = f(1+0) = \lim_{h \to 0} f(1+h) = \lim_{h \to 0} [1+h] = 1$

$LHL = f(1-0) = \lim_{h \to 0} f(1-h) = \lim_{h \to 0} [1-h] = 0$

$RHL \neq LHL$

$|f(x)| \to |l|$ converse not true.

$f(x) = \begin{cases} 1 & x < a \\ -1 & x \geq a \end{cases}$ $\qquad RHL \neq LHL$ lim. doesn't exist

$|f(x)| = 1$

① $\lim_{x \to 0} x^{\alpha} \sin \frac{1}{x^{\beta}} = 0 \qquad \alpha > 0$

$x \to 0$ $\qquad$ x^{β}

(2) $\lim_{x \to 0} x^{\alpha} \cos \frac{1}{x^{\beta}} = 0 \quad \alpha > 0$

(3) ~~lim~~ If $|f(x)| \leq K|x| \quad \forall x$

$\lim_{x \to 0} f(x) = 0 \qquad \frac{\sin x}{x} \leq 1$

Eg: $f(x) = \sin x \sin \frac{1}{x}$

$|f(x)| = |\sin x \sin \frac{1}{x}| \leq |\sin x|$

$\leq |x| \quad K = 1$

$\lim_{x \to 0} f(x) = 0$